

Practical 10 – Part 2

Unsupervised Learning

In this tutorial, you will work with Unsupervised Machine Learning, specifically, with K-Means Clustering Algorithm.

- You will perform k-means manually (to understand how it works) to group data points into clusters.
- You will then use scikit-learn library to run K-Means clustering algorithm in Python.
- You will need Jupyter notebook for Python 3.

Exercise 1: K-Means clustering algorithm manually

Consider the coordinates of the following points.

P1 (2, 10)

P2 (2, 5)

P3 (8, 4)

P4 (5, 8)

P5 (7, 5)

P6 (6, 4)

P7 (1, 2)

P8 (4, 9)

Task 1. Using pen and paper you are required to perform k-means clustering, with $k=3$, determining which points belong to which of the three clusters. To do so:

- i. Use the three points: (2,10), (5,8), (1,2) as initial cluster centers.
- ii. Show all iterations (**remember to stop when the means are not changing anymore**) of the K-Means clustering algorithm. Additionally, at each iteration you must indicate which points belong to each cluster and the coordinates of the three new cluster centers.
- iii. For each iteration you need to show graphically the points, and the mean that belong to each cluster.
- iv. The distance between each pair of points must be calculated using Euclidean metric.

To calculate the distance, you can use one the following links:

<https://www.calculatorsoup.com/calculators/geometry-plane/distance-two-points.php>

<https://www.translatorscafe.com/unit-converter/en-US/calculator/two-points-distance/>

Exercise 2: K-Means clustering algorithm in Python (1)

We are going to run K-Means Algorithm to cluster our data using Python. Follow the steps:

Step 1. Open Jupyter

Step 2. Import the following libraries:

```
import numpy as np

from sklearn.datasets import make_blobs

from sklearn.cluster import KMeans
```

Step 3. Next, we need data. We will work with some random coordinates (let us go with 50), with k=2. The initial cluster means will be [(10,15), (4,7)]:

```
num_of_coordinates = 50

initial_cluster_means = [(10,15),(4,7)]

number_of_clusters = len(initial_cluster_means)
```

Step 4. We generate the data:

```
dataset=make_blobs(n_samples=num_of_coordinates,centers=initial_cluster_means,n_features=2,cluster_std=2)
```

Check more about *make_blobs()*:

https://scikit-learn.org/stable/modules/generated/sklearn.datasets.make_blobs.html

Step 5. Display the data (50 random coordinates):

```
random_data=dataset[0]

print(random_data)
```

Step 6. Create kmeans object:

```
k=KMeans(n_clusters=number_of_clusters)

k
```

Check more about *KMeans()*:

<https://scikit-learn.org/stable/modules/generated/sklearn.cluster.KMeans.html>

Step 7. We fit the k means object to the dataset and predict. Specifically, K-Means Algorithm is run on the 50 random coordinates, and group them in their respective cluster. We can now find out the coordinates that belong to each cluster.

```
cluster_data = k.fit_predict(random_data)

cluster_data
```

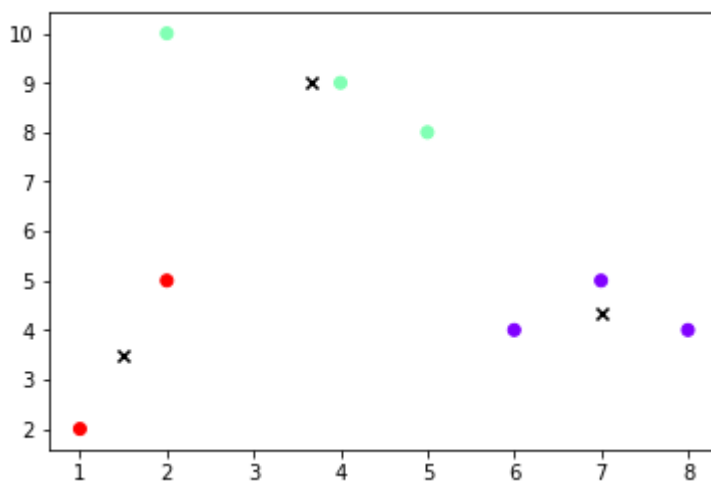
Step 8. With the knowledge that you have gained earlier, do you think you can show graphically the points and the mean that belong to each cluster? Give it a go. If you are struggling ask the teaching staff for help.

Exercise 3: K-Means clustering algorithm in Python (2)

We are now going to identify the K-Means clusters from the data given in Exercise 1 using python.

Task 1. You are required to run basic K-Means Algorithm in python to group the data points (those given in Exercise 1) into three clusters. Also, you will show graphically, the points, including the mean that belong to each cluster. You will need the code given in Exercise 2 to help you complete this exercise. However, there will be some minor changes in your code. For example, in Exercise 1 we gave K-Means the initial cluster means, but here we will let K-means to randomly pick three data points as the initial cluster means. Note that when you run basic K-Means only the last iteration, including the final clusters with their points and mean will be executed. Check if the last iteration that you had in Exercise 1 is the same when you run K-Means in Python. Give it a go. If you are struggling ask the teaching staff for help.

The generated clusters should look as shown below.



Summary

In this tutorial with regards to unsupervised learning, you have learned how to use K-Means Algorithm to cluster data. You have also gained knowledge of performing K-Means clustering algorithm using scikit-learn library in Python.

Reading

Textbook: Elements of Statistical Learning

- <https://web.stanford.edu/~hastie/Papers/ESLIII.pdf>
- Chapter: 13
- Chapter: 14, Sections (14.3.6-14.3.7)